

CHAPTER 2: RESEARCH PROCESS

Dr. Muhammad Ahmad Niaz
Assistant Professor,
Department of Computer Science,
IUB, Bahawalnagar campus.



OUTLINE



IDENTIFICATION OF
PROBLEM AREA



STATE RESEARCH
OBJECTIVE



IDENTIFYING
RELEVANT VARIABLE



HYPOTHESIS
FORMULATION



DATA COLLECTION
METHODS



DATA ANALYSIS



INTERPRETATION OF
DATA

CHAPTER OBJECTIVES

1

Describe the research process from formulating questions to seeking and finding solutions.

2

Describe the difference between dependent and independent variables.

3

Identify other types of variables that may interfere with the research process.

4

Define a hypothesis and describe how it works.

5

Discuss the value of the null hypothesis.

CHAPTER OBJECTIVES - CONTINUED

1

Describe the differences between a null hypothesis and a research hypothesis.

2

List the characteristics of a good hypothesis.

3

Explain the difference between a sample and the population.

4

Define statistical significance and explain its importance.



WHAT IS RESEARCH ALL ABOUT, ANYWAY?

Increasing our
understanding of
how and why we
behave the way we
do!!!

THE RESEARCH PROCESS: COMING TO TERMS



From Problem to Solution

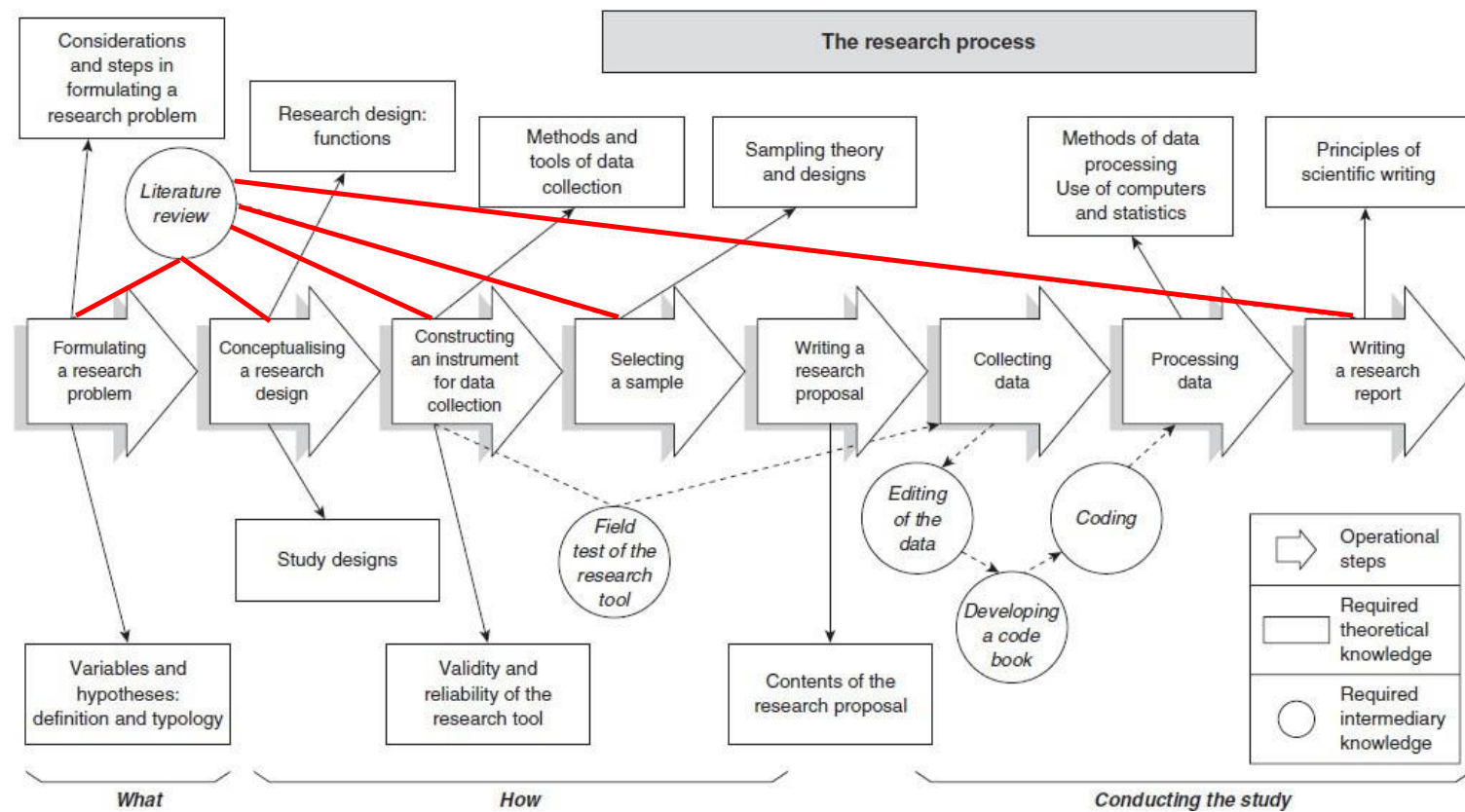
Noting an interesting question

Stating the question in such a way that it can be answered

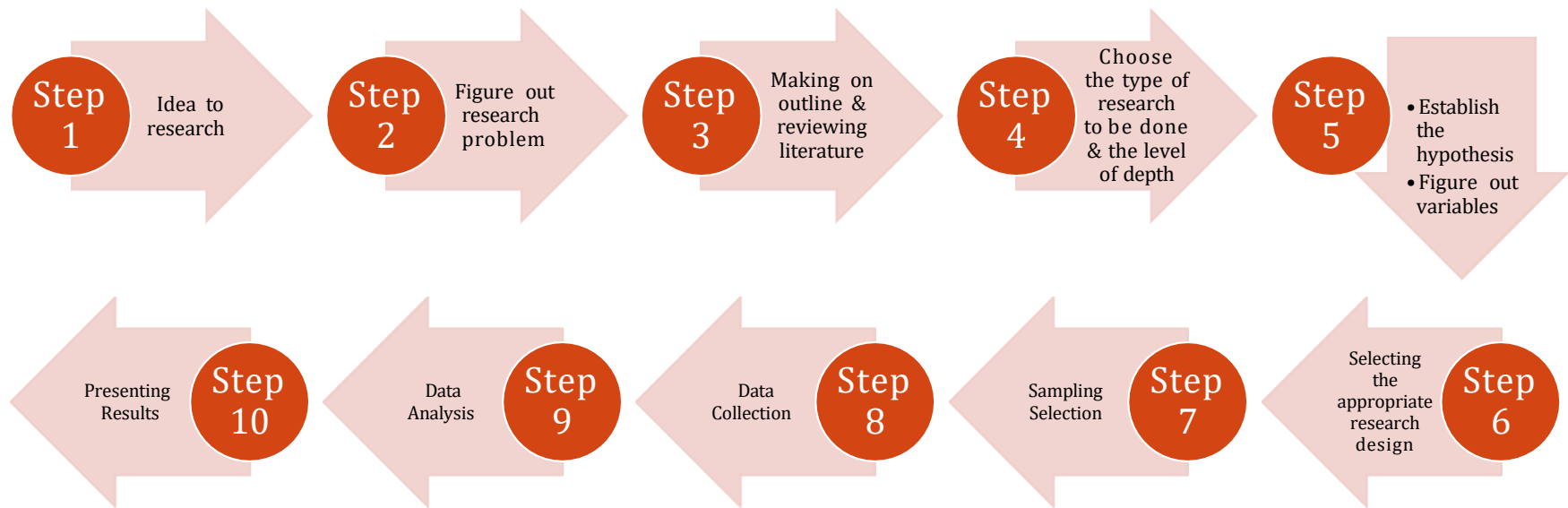


The Language of Research

Research process (Ranjit Kumar, 2018)

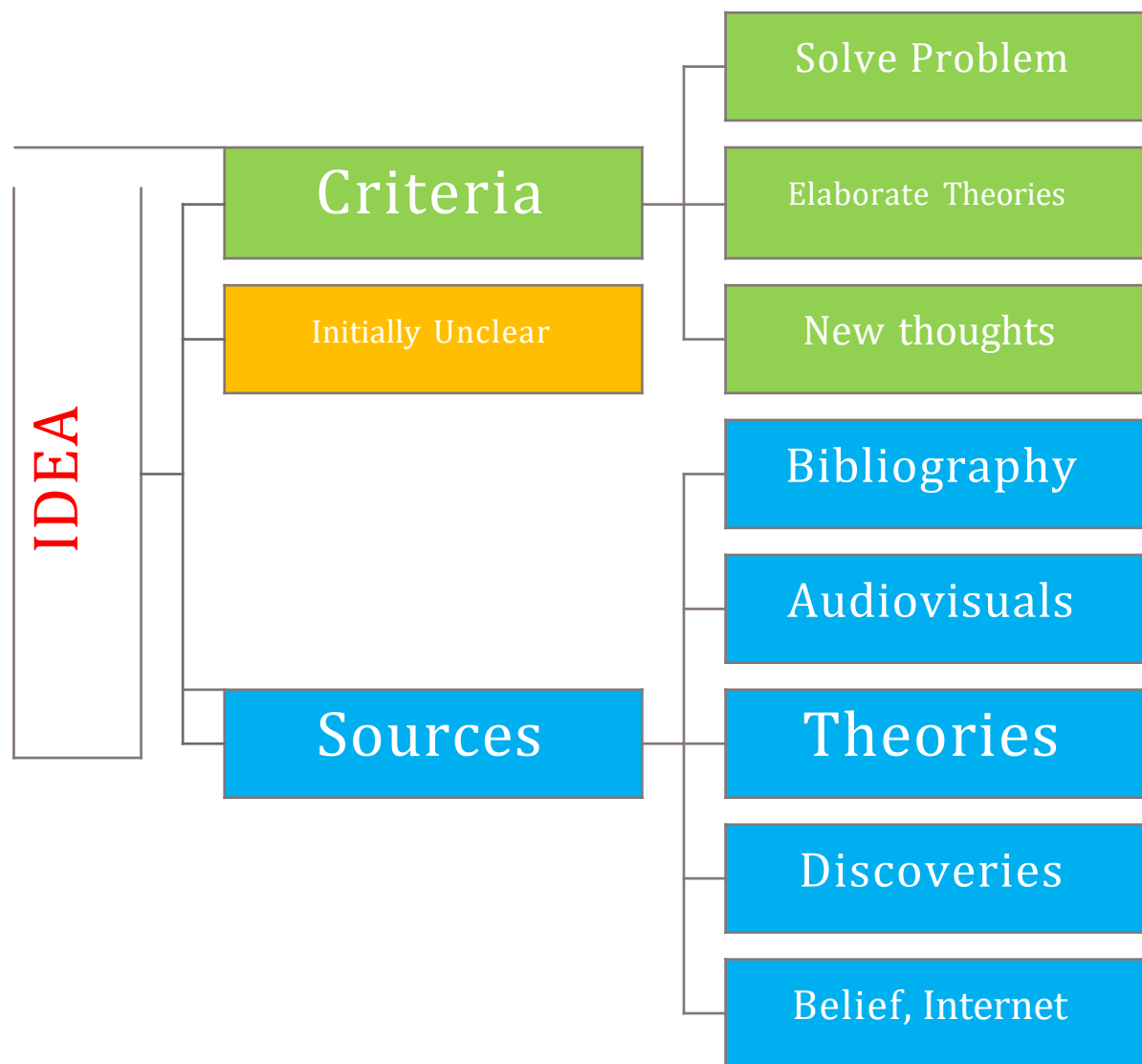


RESEARCH PROCESS





IDEA TO RESEARCH





IDEA

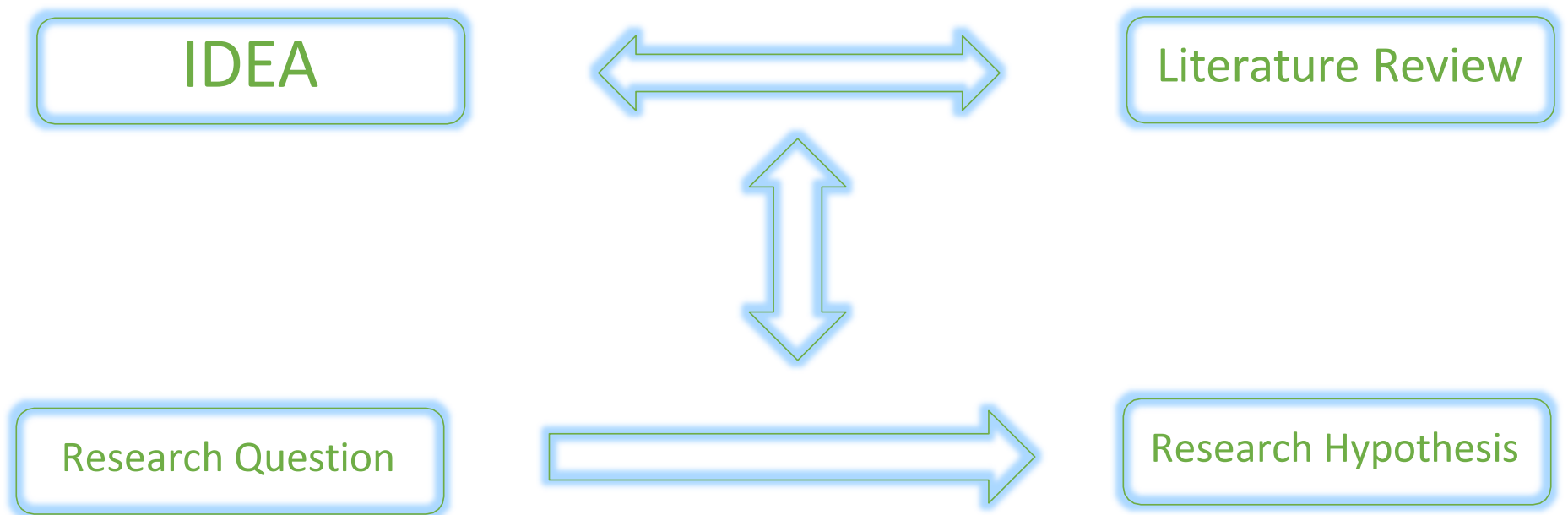
- An idea represents the closest APPROACH to reality that will be researched.
- There are lots of different resources that generate ideas
- In the beginning ideas are **UNCLEAR**. They require careful analysis in order to transform into structured and precise research statement.



IDEA TO RESEARCH: EXAMPLE

- In order to decide the researcher will have to read **articles, book, talk to specialized professionals, talk to actual couples.**
- All of this to get better **FAMILIARIZATION** with the topic and be able to select a specific topic.

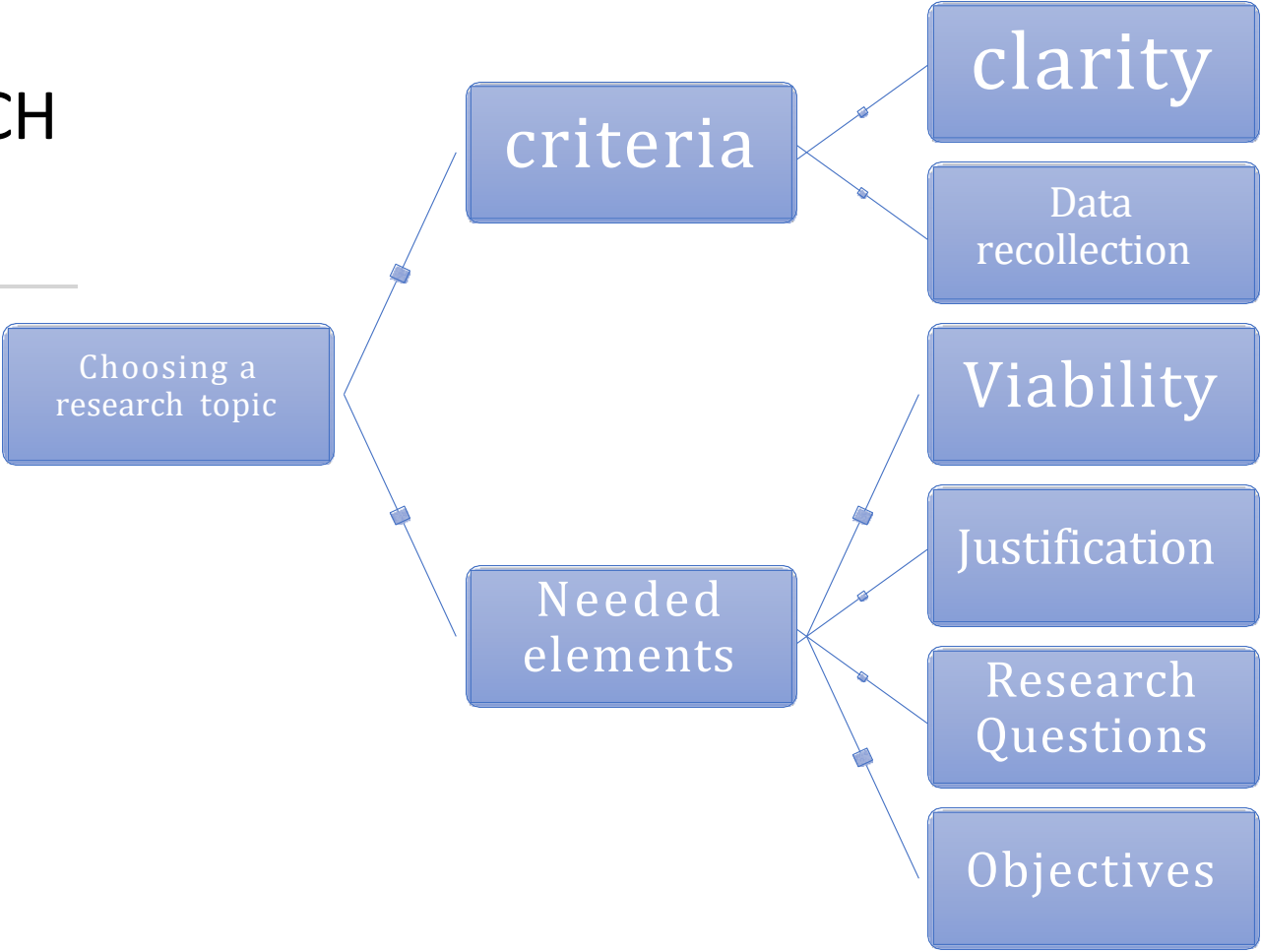
What next?



20 ways of generating research ideas from previous research

Repeat studies/replication	Do a study suggested by the journal article’s author(s).
Repeat the study with a different group of participants or under different settings.	Look for situational factors that may moderate the effects.
Look for factors that were not controlled.	Reduce the effects of expectancies.
Use more realistic amounts of the treatment factor.	Uncover the functional relationship.
Use more realistic stimulus materials.	See if another factor would have the same effect.
Bridge fields and try to find a practical implication of the research.	Look at the studies from a different level of analysis.
Look for patterns in conflicting studies.	Look for a factor’s immediate relationship to other variables.
Look at long-term effects.	Look for “down the road” effects.
Repeat the study using a different measure of the same construct.	Repeat the study with a more sensitive way of detecting the effect.
Take advantage of “component” measures.	Take advantage of measure of entirely new concepts.

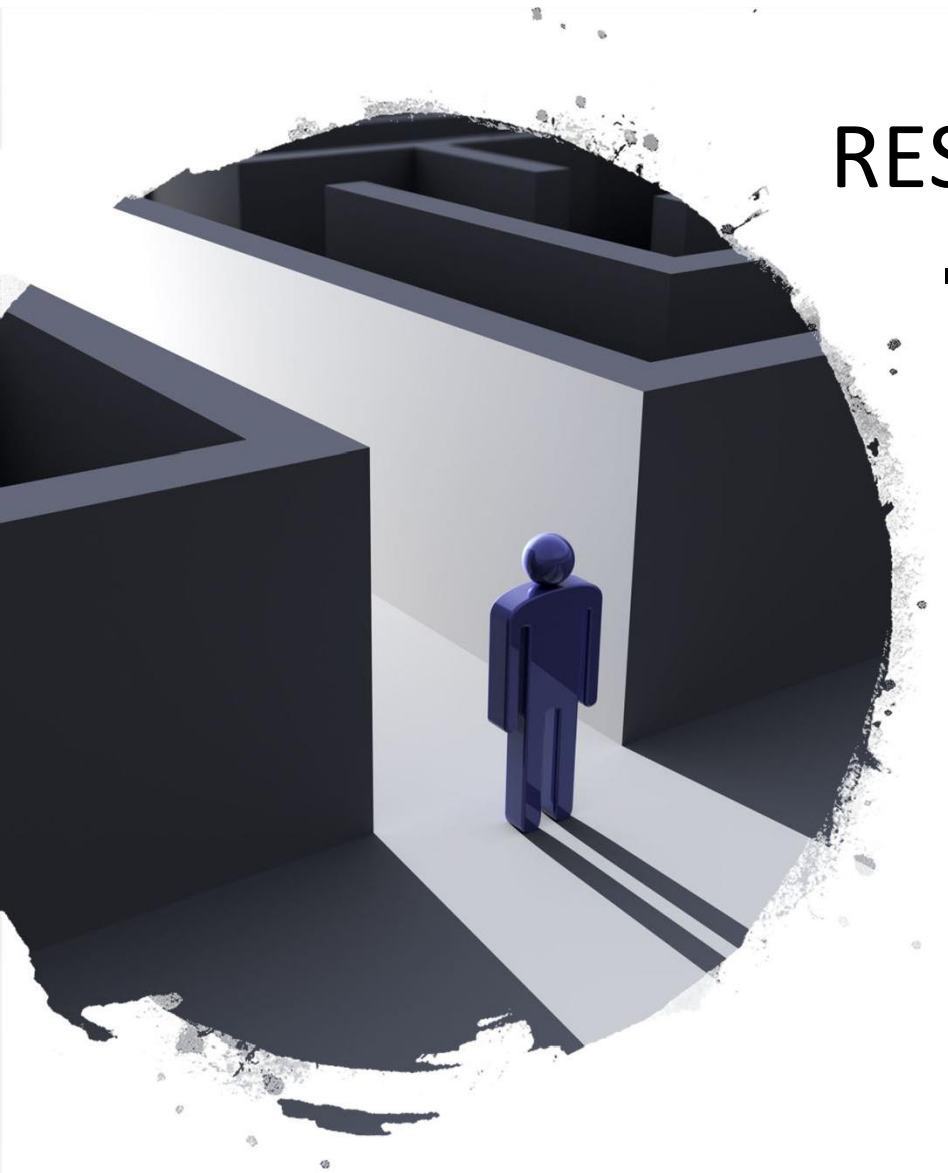
CHOOSING A RESEARCH TOPIC



RESEARCH TOPIC



- Once the researcher got acquainted and familiar with the topic then **it is possible to setup the research topic.**
- There is certain CRITERIA that is needed in order to narrow down the general topic into the “Just Right” (Research) topic.



RESEARCH TOPIC

- The criteria:
 - Has to be straight forward, usually as a question
 - Should be viable and verifiable by empirical proof
 - For example
 - “How Inspirational is a person’s soul?”
 - This is not a good research topic since inspirational is not something that is **MEASURABLE**, neither is a person’s soul.

RESEARCH TOPIC

- There are also some elements that should be taken into consideration in order to choose a research topic.
 - Objective: Guideline for the research
 - What is the main PURPOSE for the research?
 - Solve problem
 - Prove a theory
 - Introduce new knowledge

RESEARCH TOPIC

- To analyse if gender matters in the level of importance that is assigned to the different factors
- To analyse if age matters in the level of importance that is assigned to the different factors.

RESEARCH TOPIC

- Questions: must be together with the objectives, creating question will present the topic in a direct manner, minimizing the distortion
- The question or questions might not deal with the topic in its entirety, but they will also **help guide the research**.

RESEARCH TOPIC

- Questions should be concrete, since general questions will not help.
- Questions like:
 - why are some people happier than others?
 - Why do some marriages last more than others?
 - How does media relate to people's voting?
- THIS QUESTIONS ARE TOO GENERAL AND ACTUALLY END UP GENERATING MORE QUESTIONS!!!

RESEARCH TOPIC

- JUSTIFICATION: This is the explanation for your research, and it's importance.
 - Some of the reasons why a specific research might be important:
 - CONVENIENCE: Why is its importance?
 - SOCIAL RELEVANCE: What benefits would come out of it?
 - PRACTICAL IMPLICATIONS : Is it Practical?
 - THEORETICAL VALUE

RESEARCH TOPIC

- VIABILITY: this has to do with the possibility of performing research with the resources that are available.
- Do you have enough money/time/people to help you carry on the research?





RESEARCH TOPIC: EXAMPLE

- Topic: Children and TV
- Objectives:
 - To investigate the time that children dedicate to watch television.
 - To describe which shows are most watched by children who live in Shah Alam.



RESEARCH TOPIC: EXAMPLE - CONTINUE

- Objectives: - cont
 - To determine the functions and satisfactions that children, who live in Shah Alam, obtain by watching TV.
 - To determine the amount of control that parents have over their children TV watching activity.
 - To analyse what type of children watch the most TV.

RESEARCH TOPIC: EXAMPLE

- Research Questions:
 - How much time do different types of children spend watching TV?
 - What are these children's favorite TV show?
 - What type and how much control do parents have over how much TV their children watch?



RESEARCH TOPIC: EXAMPLE

■ JUSTIFICATION

- For most children, watching TV, Sleeping and going to school constitute their main activities.
- Moreover, TV is the favourite mass media technology used by smaller children.
- It is estimated than, in average, a child watched 4 hours of TV daily,
- Thus, it was calculated that by the time the child reaches the age of 15, a teenager has watched over 15,000 hours of television.

RESEARCH TOPIC: EXAMPLE

- JUSTIFICATION

- This fact has generated question **amongst parents, teacher, researcher, and others** about the relationship between a child and television and the effects that this has over the child.
- Therefore it is important to study the relationship by analysing the influence that the television exerts over children. The outcome of this study will provide **teachers and parents** on the important information about a better way to manage the relationship between children an television.

RELATIONSHIP BETWEEN RESEARCH OBJECTIVES, QUESTIONS AND HYPOTHESES



Research objectives, questions and hypotheses are **inter-related**.



Research hypotheses are possible answers to the research questions



Objectives are more general in nature, while research questions and hypotheses are more specific



SAMPLE RESEARCH OBJECTIVES

1. To measure the effectiveness of a road safety campaign
 2. To identify factors which affect palm oil prices
 3. To compare the performance of full-time and part-time students of the MBA program
 4. To find the relationship between level of stress and job performance among government doctors.
 5. To measure the satisfaction level of travellers toward services offered at KLIA
 6. To evaluate the success of the privatization project of property development at the local government level.
-

SAMPLE RESEARCH QUESTIONS

(1)



Objective (1) To measure the effectiveness of a road safety campaign

- 1) Are road users aware of the campaign?
- 2) What is the attitude of road users towards the campaign?
- 3) Is there a significant reduction in the number of accident cases as a result of the campaign?

Objective (2) To identify factors which affect palm oil prices

- 1) Is the price of palm oil affected by the price of other vegetable oils?
- 2) Which is the most important factor that affect palm oil price?
- 3) What is the impact of an increase in the demand of soy bean oil on the price of palm oil?

SAMPLE RESEARCH QUESTIONS (1)

1. Psychology:

- a) Do women tend to perceive other women as less competent than men as bosses?
- b) What are the effects of no-smoking policy to absenteeism and morale of employees?

2. Business:

- a) Does advertising reach the right people?
- b) Does giving frequent special discounts eventually lead to reduced sales in the long run?

3. Biology:

- a) Why do birds fly in a formation?
- b) Are rats smarter than frogs?

SAMPLE RESEARCH QUESTIONS (2)

Anthropology:

- a) Is there evidence of differences in skeletal heights of prehistoric tribes?

Medicine:

- a) How much lead is consumed daily through our drinking water?
- b) What biological behavioral and socio-structural factors contribute to oral disease?

Law:

- a) Is there gender/ethnic bias in jury selection?



null hypothesis $\rightarrow H_0$: There are no relationship between A and B.

alternative hypothesis $\rightarrow H_1/H_A$: There is a relationship between A and B.

hypothesis

- An educated guess or prediction of what will happen.
- Proposes a relationship between factors called variables.
- A good hypothesis relates an independent variable and a dependent variable.
- A good hypothesis is one you can test using the scientific method.

Key points of hypothesis



Does the hypothesis relate an independent and dependent variable? Can you identify the variables?



Can you test the hypothesis? In other words, could you design an experiment that would allow you to establish or disprove a relationship between the variables?



Would your experiment be safe and ethical?



Is there a simpler or more precise way to state the hypothesis? If so, rewrite it.

Research Question

- 1) What is the relationship between no of cigarette smoke and Risk of Heart disease?

Research Objective

- 1) To measure the relationship between no of cigs smoke and Risk of heart disease.

Research Hypothesis

- 1) There is ^a relationship between no of cigarette smoke and the risk of heart disease.

analysis.

IV

V₁

V₂

DV

SAMPLE NULL HYPOTHESIS

- There is **no significant difference** in the level of job satisfaction between male and female lecturers.
- There is **no significant effect** of stress on job performance of teachers
- There is **no evidence** of gender bias in the hiring of new graduates by the private sector
- There is **no significant impact** of SARS on tourism in South East Asia
- There is **no evidence** of dissatisfaction among tourists towards the taxi service at KLIA
- There is **no significant** relationship between satisfaction and duration of stay among foreign tourists.

IDENTIFY THE DEPENDENT AND INDEPENDENT VARIABLES FOR EACH RESEARCH QUESTION BELOW

1. Which teaching approach had the greatest appeal to primary school students?
2. Is there a relationship between the **occurrence of backache** and **the amount of time spent** in front of computers?
3. Are female graduates being **discriminated** in the job market?
4. How does fruit juice consumption affect growth parameters during early childhood?
5. Is Islamic banking well understood by the masses?
6. Which services offered at KLIA are below expectation?
7. Which is the most popular TV station in Malaysia?
8. Are there any extraneous and moderator variables?



EXERCISE

- State problem statement, research objective(s), and research question(s) for the following situation.

SITUATION 1

- Baitulmal College at Kota Bharu is planning to introduce a free bus service within the campus for their lecturers, administrative staff, students and visitors. This is one of the several alternatives proposed to reduce traffic congestion within the campus. The administrative staff, students and visitors who are interested to use the services. Baitulmal College would also like to identify strategic location outside the campus as pick-up points and also interested to determine frequency of pick-ups for each point.





SITUATION 2

- Customer complaints were mounting, and letters of complaint regarding the problems they experienced with the residential telephone lines were constantly pouring in at the Friendly Telephone Company, a telecommunication service provider. The company wants to pinpoint the specific problems and take corrective actions and at the same time, the company also wants to investigate whether this problem has affected their customer's loyalty towards their services.
- Researchers were called in, and they spoke to a number of customers, noting down the nature of the specific problems they faced. Since the problem had to be resolved very quickly, they developed a theoretical base, collected relevant information from a sample of 100 customers, and analysed the data. The results are expected to be fairly accurate with at least an 85% chance of success in problem solving. The researchers would make recommendations to the company based on the results of the data analysis.

SITUATION 3

- Stress at work has undesirable consequences on the health and safety of the employee and their organizations. Research conducted in European countries found evidence of the effectiveness of management intervention to reduce workers' stress. Poor organization cultures such as long working hours and uncondusive working environment is associated with an increased experience of stress. Effective stress management programmes have been found to reduce the effects of stress on health.
- The management of a company conducted a study on work stress at work among its employee. The study aims to determine how stress affects the health and well being of their employee and its implication on the productivity of the company. It is hoped that the results of the study will assist management to deal with stress at work among the employees, thus boosting the productivity of their employees. A random sample of 200 workers was selected for the study.

SITUATION 4

- One of the factors that contribute to the robustness of the Malaysian economy is the tourism industry. The tourism industry has a potential to assist the Malaysian economic expansion. However, it mostly depends on the number of tourist arrivals in the country. Taman Tamadun Islam (TTI) is one of the theme park that places a gorgeous and historic monument from all Islamic worlds, not just a new symbol for Terengganu but Malaysia as well.
- As found from the other studies, the tourists intend to visit if they satisfied with the service provided at the visited place. To attract more tourist come to the Malaysia, some of the critical issues that believed need to be concern are the price, food, cleanliness of surrounding, public transport condition, safety of the tourist and also quality of services provided. Besides, demographic profiles such as gender, age, visitor country and etc, found to be significantly influence visitors' satisfaction. Therefore this study intends to determine the factors contribute to the visitor's satisfaction.
- After reviewing the literature, the researcher found that tourist satisfaction can be determined as the results of the difference between the expectation and perception. It is also claimed that quality of services and tourist satisfaction develop long-term relationship with visitors and in turn bring about destination loyalty. Apart from that, this study also wants to determine whether there exist relationship between visitors' satisfaction and their loyalty.